CS 4400: Artificial Intelligence

Probability and Bayes' rule 1

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1/8. Weakness of logic

- Logic is very good at describing definites
  - Blue(c1) ∧Car(c1) : "c1 is a blue car"
  - ∀xExam(x) ⇒~Like(Me,x)
- But not very good at indefinites
  - KB: 50-years of records in humidity, temperature, etc
  - ASK: Will it rain tomorrow?
    - Cannot say with certainty that it will or will not rain tomorrow…
    - There's a 10% chance of rain tomorrow

Automated taxi

- Goal: deliver a passenger to airport (5-miles away) on time
- Plan A_{90}: leave home 90 mins before departure and drive at a reasonable speed
- ASK: Is the plan guaranteed to work?
  - The plan A_{90} will work, if the car doesn't break down, run out of gas, nor get into an accident, and if there is no accident on the bridge, no meteorite hits the car, no earthquake on the road, …
  - Success cannot be inferred since we cannot rule out any of these occurring for sure, however unlikely these events are

Uncertainty

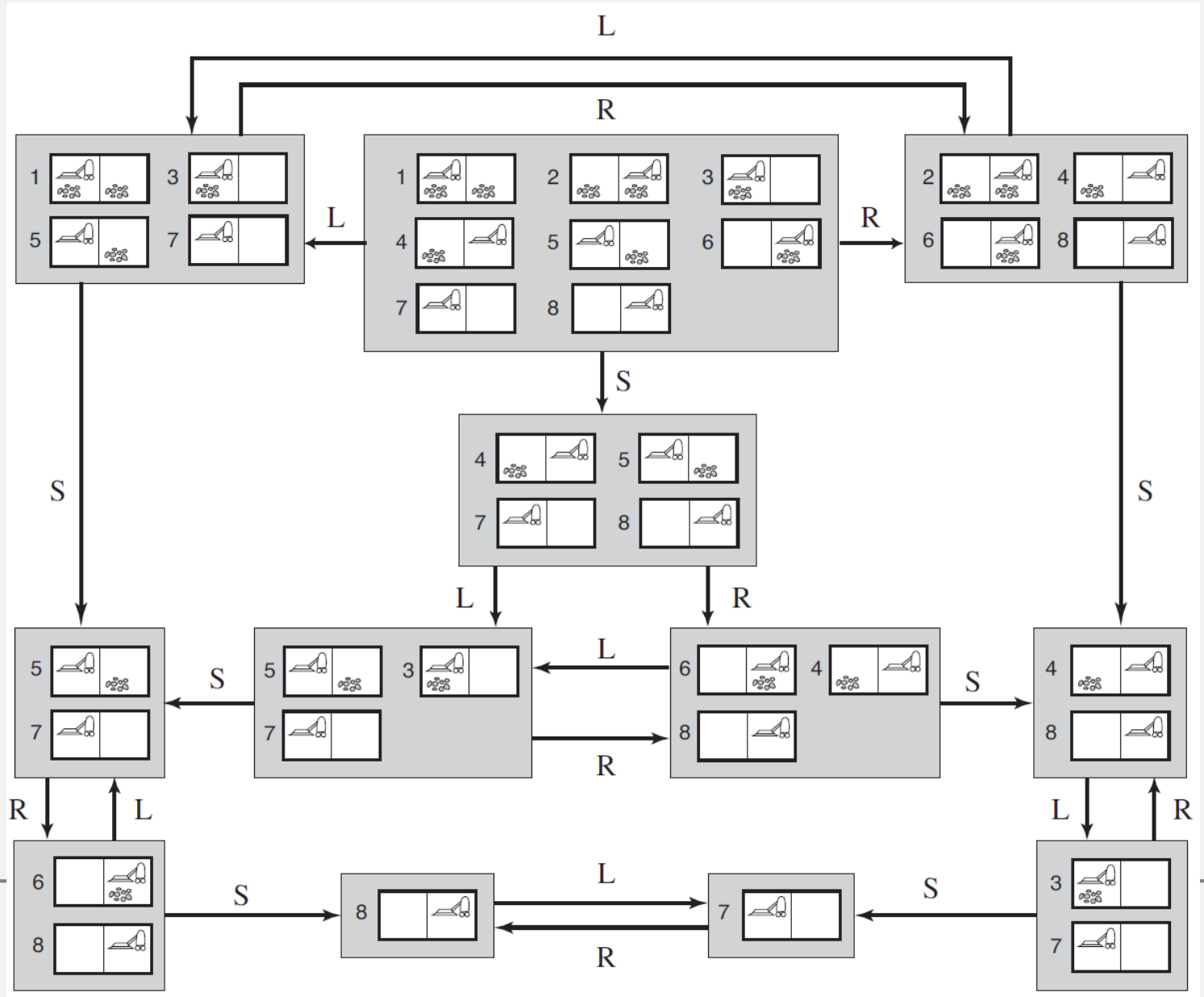
- In the real world, uncertainty is everywhere
- In vacuum world
  - Noisy (=unreliable) sensors
  - Uncertainty in action outcomes (erratic vacuum, slippery vacuum)
  - Uncertain environment (kindergarten rule)
  - Partial observability (local sensing vacuum, driving)

Suck at square A
- When A has dirt, it removes dirt, sometimes removes dirt in B as well
- When A is clean, it sometimes deposits dirt

Movement actions sometimes fail, leaving the agent in the same location.

- In search, uncertainty was handled by belief state
  - Belief states: any possible subset of the powerset: 2^N
  - E.g., states = {1}, {2}, belief state = {1}, {1,3,4}

Belief state-space (deterministic, sensorless)



Problems of using belief states

- Belief state space can be too large
  - Need to consider every possible cases, no matter how unlikely
- Belief state space may not guarantee that the goal is reached
  - Yet the agent must act
- No way to compare merits of different plans
  - Plan 1: use interstate 1, which may have heavy traffic during commuting hours
  - Plan 2: use local, which is slower but never gets congested

Medical diagnosis example

- Using logic to explain relationships between Toothache, Cavity, etc…
- Toothache ⇒Cavity ?
  - Not all toothaches are caused by cavities…
- Toothache ⇒ Cavity v GumProblem v Abscess …
  - The list never stops
- Cavity ⇒Toothache ?
  - Not all cavity cause pain.
- Uncertainty is inherent, because
  - We cannot list all possible conditions
  - We do not have complete theory (e.g., of medical diagnosis)
  - We cannot have all information (e.g., from medical tests)

Probabilistic thinking

- Probabilities = degree of belief in a particular proposition
  - No longer just true or false
  - "The chance of dying after a car accident is 0.01%"
    - P("Car accident ⇒Death") = 0.0001
    - It does not mean someone can be dead with 0.0001 intensity
    - The person will or will not die, with 99.99% chance of living
  - "Toothache patients have cavities 80% of the time"
    - P("Toothache ⇒Cavity") = 0.80
    - It does not mean that every patient has 80% cavity
    - Each patient has or does not have cavity, with 80% chance of cavity

Unconditional probabilities

- Unconditional (or prior) probabilities: the usual or regular probabilities we know of
- Gives the probability of a variable taking a value without any conditions on other variables
  - P(Roll=2) = 1/6
  - P(Roll >=4) =1/2
  - P(rainToday) = 0.10
  - P(~rainToday) = 1 – 0.10 = 0.90
- Cf. Conditional (or posterior) probabilities

2/8. Formal definitions

- Begin with a set Ω called the sample space (=universe)
  - An element ω is a sample = a possible world = an atomic event
  - e.g., A domain with a single die: Ω = 6 possible rolls
- A probability space (or model) is a sample space with an assignment P(ω) for every $\omega \in \Omega$ s.t.
  - $0 \le P(\omega) \le 1$, $\sum_{\omega} P(\omega) = 1$
  - e.g., for an unbiased die roll, P(Roll=1)= … =P(Roll=6)=1/6, P(Roll=1) + … + P(Roll=6) = 1
- An (possibly non-atomic) event A is any subset of Ω
  - $P(A) = \sum_{\{\omega \in A\}} P(\omega)$
  - e.g., A: Roll<4, P(A) = P(Roll=1) + P(Roll=2) + P(Roll=3) = 3*1/6 = 1/2

Random Variables

- We need symbols to refer to events: random variables
  - E.g., Cavity : {True, False} or {1, 0}
- For binary (true or false) random variables, we abbreviate:
  - “variable1” iff “Variable1=True”
  - “~variable1” iff “Variable1=False”
- A random variable can be multi-valued
  - Roll = number from a roll of a die: {1,2,3,4,5,6}

Probability distribution function

- Probability distribution gives values for any possible assignments of random variables:
  - Suppose Weather is one of {sunny,rain,cloudy,snow}
  - P(Weather=sunny) = 0.72, P(Weather=rain) = 0.1, P(Weather=cloudy) = 0.08, P(Weather=snow) = 0.1
  - P(Weather) = <0.72, 0.1, 0.08, 0.1>
- Consider this example:
  - Temperature: a real value between 18C – 26C
  - P(Temperature) = 1/8, if 18 ≤ Temperate ≤ 26

    0, otherwise

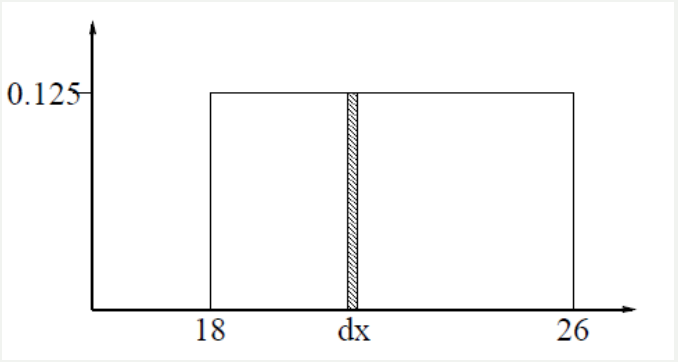
    that is, the Temperature has a uniform distribution

Probability density function

- What is the probability that temperature=20.5?
  - According to the previous definition, P(20.5) = 1/8?
- For continuous variables
  - P(x) ≠ probability of x

$$P(x) = \lim_{dx \to 0} P(x \le X \le x + dx)/dx$$

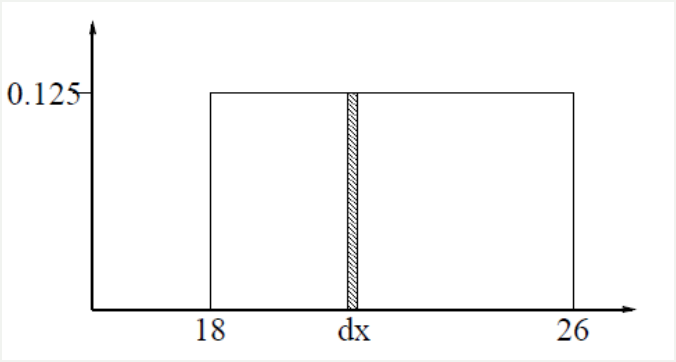
$$\lim_{dx \to 0} P(20.5 \le X \le 20.5 + dx)/dx = 0.125$$



  - It's sometimes called probability density function
  - Practically, you should interpret P(x) differently depending on whether x is continuous or discrete

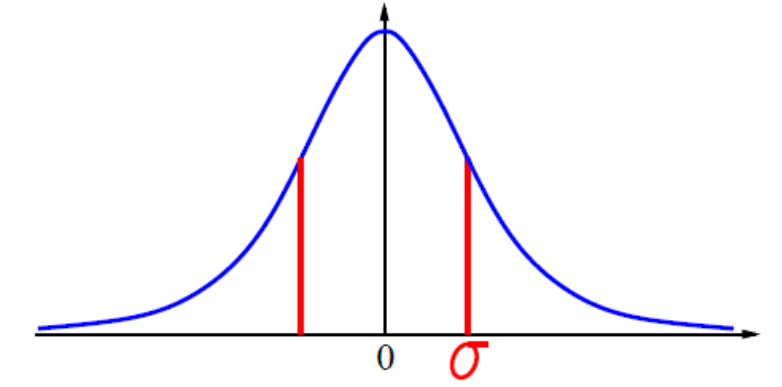
Probability density function

- Uniform density function



- Gaussian density function (a.k.a. Normal density function)

$$P(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$



Multiple Variables

- Problems can have multiple variables in the probability space/model
  - If I have a toothache, do I have a cavity? (2 variables)
  - How do height and weight differ by gender? (3 vars)
  - What's the probability of getting to 21 in Blackjack in two cards? (2 vars)
    - If I have less than 21, what's the probability of hitting 21 in 3 cards (3 vars.)
- How do we relate these multiple variables?

3/8. Joint Probability Table

- We can use a Joint Probability Table (JPT) to specify the joint probability of multiple variables
  - Examples:
    - P(rainToday ∧windToday) =P(rainToday, windToday) ?
    - P(~rainToday ∧windToday) =P(~rainToday, windToday) ?
    - Note: When variables are binary, we can use logical connectives

	rainToday	~rainToday
windToday	0.08	0.2
~windToday	0.02	0.7

Marginalization

	rainToday	~rainToday
windToday	0.08	0.2
~windToday	0.02	0.7

- From JPT, we can calculate the probability of individual variables by summing the columns or rows
- It's called marginalization
  - P(windToday) = 0.28
  - P(~rainToday) = 0.9

Conditional Probabilities

- If we know the values of some variables, how does it affect the probability of other variables?
- Conditional (or posterior) probability
  - P(A=a | B=b): probability of event A=a given that we know event B=b
- Examples
  - I know my brakes are squeaking, are they bad?
    - P(badBrakes | squeaking)
  - I have a toothache, do I have a cavity?
    - P(cavity | toothache)

Conditional Probabilities

- The conditional probability is defined by the joint and unconditional probabilities

$$P(A=a \mid B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$ (assuming $P(B=b) \neq 0$)

- **Product rule**: $P(A=a,B=b) = P(A=a|B=b)\ P(B=b)$
- Note: a and b are very often omitted by convention
  - $P(A|B) = P(A,B) / P(B)$, or $P(A,B) = P(A|B)\ P(B)$

Conditional Probabilities

	rainToday	~rainToday
windToday	0.08	0.2
~windToday	0.02	0.7

- P(rainToday | ~windToday)?
  - P(rainToday,~windToday) / P(~windToday)
  - 0.02 / (0.02+0.7) = 0.02/ 0.72 ~= 0.027
  - Cf: P(rainToday) = 0.1

Conditional Probabilities

	rainToday	~rainToday
windToday	0.08	0.2
~windToday	0.02	0.7

- P(rainToday | windToday)?
  - P(rainToday,windToday) / P(windToday)
  - 0.08 / 0.28 ~= 0.285
- Important !
  - P(a) + P(~a) = 1, P(a|b) + P(~a|b) = 1, proof: straightforward
  - but not necessarily P(a|b) + P(a|~b) ~=1

4/8. Inference with JPT

- Given a JPT of multiple random variables, we can answer most of the questions using
  - Marginalized probabilities
  - Conditional probabilities
- Inference by enumeration
  - C.f. inference by enumeration in propositional logic
- Examples to follow …

Inference Example

- Three variables: Cavity, Toothache, Catch (Catch is true if the dentist's steel probe catches in the cavity)
- What is P(cavity) from this JPT?
  - Sum the entries where Cavity is true

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity) from this JPT?
  - Sum the entries where Cavity is true

    P(cavity) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity ∨ toothache) from this JPT?
  - Sum the entries where either Cavity or Toothache is true

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity ∨ toothache) from this JPT?
  - Sum the entries where either Cavity or Toothache is true
  - 0.108+0.012+0.016+0.064+0.072+0.008 = 0.28

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity $\vee$ toothache) from this JPT?
  - Use the rule of disjunction
  - P(cavity $\vee$ toothache) = P(cavity) + P(toothache) – P(cavity $\wedge$ toothache)

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity $\vee$toothache) from this JPT?
  - Use the **rule of disjunction** : P(a $\vee$b) =P(a) +P(b) – P(a $\wedge$b)
  - P(cavity $\vee$toothache) =P(cavity) +P(toothache) – P(cavity $\wedge$ toothache) =0.28

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity $\wedge$toothache) from this JPT?
  - Sum the entries where both Cavity and Toothache are true

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Inference Example

- What is P(cavity $\wedge$toothache) from this JPT?
  - Sum the entries where both Cavity and Toothache are true: 0.108+0.012 = 0.12

    = $\sum_{c \in \{true, false\}}$ P(cavity,toothache,Catch=c)

	toothache		~toothache	
	catch	~catch	catch	~catch
cavity	0.108	0.012	0.072	0.008
~cavity	0.016	0.064	0.144	0.576

Marginalization revisited

- In general, if you have P(X,Y,Z) and you want P(X,Y)
  - Sum P(X,Y,Z) over all possible values for Z
  - $P(X,Y) = \sum_z P(X,Y,Z=z)$
    $= P(X,Y,Z=z_1) + \ldots + P(X,Y,Z=z_n)$
- Similarly, if you have P(X,Y|Z) and you want P(X|Z)
  - Sum P(X,Y|Z) over all possible values for Y
  - $P(X|Z) = \sum_y P(X,Y|Z)$
    $= P(X,Y=y_1|Z) + \ldots + P(X,Y=y_n|Z)$
- Eliminating a variable by summing over all values of that variable is called “marginalizing out.”

5/8. Bayes' Rule

- How are P(A|B) and P(B|A) related?
  - Answer: $\frac{P(A|B)P(B)}{P(A)} = P(B|A)$: Bayes' Rule
  - Proof: use definition of conditional probability
- Meaning: if you have causal model, you can use it for diagnosis, and vice versa.
  - P(Cause|Effect) = $\frac{\text{P(Effect|Cause) P(Cause)}}{\text{P(Effect)}}$

Bayes' Rule in action

- Suppose a patient came in to a clinic with a stiff neck
  - What's the probability of having a meningitis?
  - P(m|s)? (m:meningitis, s:stiff neck)
  - Could try to use the data to come up with an estimate of this directly
  - Or use other (easier) statistics and apply Bayes' rule
    - P(m) = 1/50000, P(s) = 1/20
    - P(s|m) = 0.5
    - P(m|s) = (P(s|m)P(m))/P(s) = 0.0002
- What if there were a meningitis epidemic?

Inference by enumeration

- Using the full joint probability tables for inference is straightforward, but not the most efficient
  - Cf. inference by enumeration in propositional logic
- N binary random variables: 2^N entries in the table
- Are there special relationships among variables that can simplify inferences?
  - Independence
  - Conditional independence